BAIT 508 · GROUP PROJECT

Public Firm Analysis: Oil & Gas Extraction

SIC major group 13 · Compustat panel 1994–2025 and 10-K Item 1 text, 2020 vs. 2025 · Focal firm: EOG Resources Inc.

Course: BAIT 508, UBC Sauder School of Business · **Section:** [BA2] · **Submitted:** [8 October 2026]

Team members and roles

#	Full name	Student number	Email	Role in the project
1	Anirudha Nair	68436658	anair31@student.ubc.ca	Part 2: text analysis (Steps C–F)
2	Madhav Singh	21573522	madhav20@student.ubc.ca	Part 3: firm analysis and strategy (Step G)
3	Karim Mazrou	20728531	kmazrou@student.ubc.ca	Part 1: quantitative analysis (Steps A–B) & collating analysis report

All three members reviewed each other's work and combined the parts into this single report.

Report roadmap. Part 1 A Sector selection and filtering → B Preliminary analysis · Part 2 C Text cleaning → D Keyword analysis → E Word2Vec → F 2020 vs. 2025 · Part 3 G EOG Resources: peer benchmarking and one recommendation · Conclusions, limitations, references and AI disclosure.

Key findings at a glance

- **A churning sector.** 772 oil and gas firms appear in 8,066 firm-years, but only 39 (5.1%) report in all 32 years; the number of firms per year fell from 291 (1994) to 138 (2025).
- **Prices follow the oil cycle.** The sector's average share price halved in 2008 ($24.96 → $12.98), bottomed at $12.46 in 2020 and peaked at $29.10 in 2024. The median price tells a weaker story ($13.09 in 2007, $11.04 in 2025).
- **The language is stable; the edges move.** Both years' 10-Ks centre on "gas," "oil" and "natural"; between 2020 and 2025, "drilling" falls from 45% to 35% of firms' top-10 TF-IDF keywords, while "reserves" (22% → 31%), "emissions" (6% → 14%) and completion terms such as "proppant" and "sand" gain share.
- **EOG earns more per dollar of assets than most peers.** Among the ten largest US producers, EOG Resources beat the peer-median ROA in 10 of 11 years (2015–2025) and ranked first in 2024 and 2025, on 2025 sales of $22.6 billion (6.9% of listed US producers' sales). But its ROA has fallen from 18.8% in 2022 to 9.6% in 2025.
- **Our one recommendation:** put an explicit return hurdle on balance-sheet growth. Acquisitions and major capital programmes should be expected to earn at least EOG's own 2023–25 average ROA (about 13.5%) before they are approved.

Data used

The project uses the four course files: `public_firms.csv` (Compustat-style annual panel, 247,890 firm-year rows, 1994–2025; dollar values in $ millions), `major_groups.csv` (83 SIC major groups), and the Item 1 ("Business") text of 10-K filings for 2020 (`2020_10K_item1_full.csv`, 5,481 filings, or 5,152 after removing 329 exact duplicates) and 2025 (`2025_10K_item1_full.csv`, 4,536 filings). All work was done in one Jupyter notebook (Python: pandas, scikit-learn, NLTK, gensim, wordcloud, matplotlib). Code snippets in this report are taken from that notebook.

Part 1 · Quantitative Analysis of the Industry Sector

Oil & Gas Extraction (SIC 13) · Compustat-style panel, 1994–2025

> **Pipeline.** A Sector selection and filtering (counts) → B Preliminary analysis (top prices, top sales, geography, average price, 2008 crisis, ROA)

A. Industry sector selection and data filtering

We chose **major group 13, Oil and Gas Extraction**. The sector is large, global and highly cyclical: its fortunes move with oil prices, the 2008 financial crisis, the 2014–2016 price collapse, COVID-19 and the energy transition. That makes it a good test of whether financial data and annual-report language tell the same story.

- **Filter the panel.** Compute each firm's major group as `sic // 100` (integer division) rather than slicing the SIC string: for a 3-digit code such as 100 (crops), slicing would give "10" (metal mining). Keep rows with major group 13, which covers codes 1311 (crude petroleum and natural gas), 1381 (drilling), 1382 (exploration services) and 1389 (oil and gas field services). The result is called `df_og`.
- **Count firm-years.** A firm-year is one (gvkey, fyear) pair; an `assert` confirms there are no duplicate pairs, so rows equal firm-years.
- **Count firms.** gvkey is Compustat's permanent firm identifier, so unique gvkeys give unique firms.
- **Find full-history firms.** Count the distinct fiscal years each firm reports between 1994 and 2025 and keep firms with all 32.

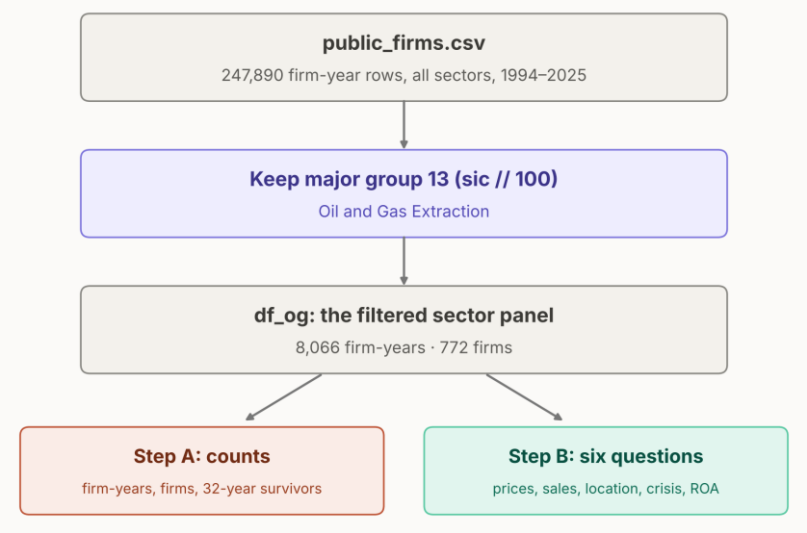


Figure 1.1. Part 1 filtering flow chart.

Code 1.1 · Filter to major group 13 and answer Step A

```
public_firms = pd.read_csv("public_firms.csv")
major_groups = pd.read_csv("major_groups.csv")

# Integer division, not string slicing: SIC 100 (crops) would become "10" (metal mining)
public_firms["major_group"] = public_firms["sic"] // 100

SECTOR_CODES = [13]  # Oil And Gas Extraction
df_og = public_firms[public_firms["major_group"].isin(SECTOR_CODES)].copy()

# (a) firm-year observations (verify gvkey-fyear is unique, so rows = firm-years)
assert not df_og.duplicated(["gvkey", "fyear"]).any()
n_firm_years = df_og[["gvkey", "fyear"]].drop_duplicates().shape[0]   # 8066

# (b) unique firms: gvkey is the stable Compustat ID; company names can change over time
n_firms = df_og["gvkey"].nunique()                                    # 772

# (c) firms present in every year 1994-2025
n_years_total = public_firms["fyear"].nunique()                       # 32
years_per_firm = df_og.groupby("gvkey")["fyear"].nunique()
full_history_ids = years_per_firm[years_per_firm == n_years_total].index
n_full_history = len(full_history_ids)                                # 39
```

Table 1.1. Step A results for the filtered oil and gas dataset.

Question	Answer
(a) Unique firm-year observations (gvkey × fyear)	8,066
(b) Unique firms (gvkey)	772
(c) Firms with records in all 32 years (1994–2025)	39

Result. The sector has high turnover. Only 39 of 772 firms (5.1%) are present for the full 32 years; the median firm appears in 8 years and a third of firms (33.5%) in five years or fewer. The number of firms reporting each year fell from 291 in 1994 to 187 in 2020 and 138 in 2025, consistent with bankruptcies, mergers and delistings after each oil-price downturn. Long-run averages in this sector therefore describe the survivors more than the typical entrant.

B. Preliminary analysis

All six questions below use the filtered dataset `df_og` from Step A. Each answer comes from a short pandas chain: filter, sort or group, and keep the top rows.

B.1 Top 10 firms by stock price, 2020 and 2025

For each year we kept the rows with that fiscal year, sorted by year-end closing price (`prcc_c`) from high to low and took the first ten.

Code 1.2 · Top 10 stock prices in 2020 and 2025

```
cols = ["conm", "prcc_c", "sale", "location"]
top_price_2020 = (df_og[df_og["fyear"] == 2020].nlargest(10, "prcc_c")[cols]
                  .reset_index(drop=True))
top_price_2025 = (df_og[df_og["fyear"] == 2025].nlargest(10, "prcc_c")[cols]
                  .reset_index(drop=True))
```

Table 1.2. Ten highest year-end stock prices (prcc_c) in the sector, 2020 and 2025.

#	2020 firm	Price ($)	2025 firm	Price ($)
1	Amen Properties	397.00	Amen Properties	505.00
2	Reserve Petroleum	142.70	Gulfport Energy	207.99
3	Pioneer Natural Resources	113.89	Cheniere Energy	194.39
4	CNOOC Ltd	91.65	Reserve Petroleum	178.63
5	Cheniere Energy	60.03	PrimeEnergy Resources	171.00
6	Nabors Industries	58.23	Diamondback Energy	150.33
7	Hess Corp	52.79	Viridien S.A.	115.57
8	EOG Resources	49.87	Expand Energy	110.36
9	Diamondback Energy	48.40	EOG Resources	105.01
10	PrimeEnergy Resources	43.17	ConocoPhillips	93.61

Result. Six names appear in both years (Amen Properties, Reserve Petroleum, Cheniere, Diamondback, EOG and PrimeEnergy), and every 2025 price in the list is higher than the 2020 cut-off, reflecting the post-COVID recovery in energy prices. Cheniere more than tripled ($60 → $194) as US LNG exports ramped up, while Pioneer Natural Resources drops out because ExxonMobil acquired it in 2024. A high share price reflects how many shares a firm has issued as much as its value: Amen Properties, a small royalty company, tops both lists.

B.2 Top 10 firms by highest annual sales, 1994–2025

We read "highest sales in the entire history" as each firm's **single best year**, and each firm appears once: we sorted all firm-years by `sale`, dropped missing values, kept the first row per gvkey and took the ten largest. Ranking raw firm-years instead would let ConocoPhillips fill 6 of the 10 slots.

Code 1.3 · Top 10 firms by highest annual sales (each firm's best year)

```
# Rank firms, not firm-years: otherwise ConocoPhillips alone fills 6 of the 10 slots
top_sales_firms = (df_og.dropna(subset=["sale"])
                   .sort_values("sale", ascending=False)
                   .drop_duplicates("gvkey")          # keep each firm's best year only
                   .head(10)[["conm", "fyear", "sale", "location"]]
                   .reset_index(drop=True))
```

Table 1.3. Ten firms with the highest annual sales in any single year (sale, 1994–2025).

#	Firm	Peak year	Peak sales ($ millions)	Location
1	ConocoPhillips	2011	230,859.0	USA
2	ENI SpA	2012	167,843.3	ITA
3	Marathon Oil Corp	2008	72,128.0	USA
4	Enterprise Products Partners LP	2022	58,186.0	USA
5	SLB Ltd	2014	48,580.0	USA
6	CNOOC Ltd	2013	47,220.2	HKG
7	Dynegy Inc	2001	42,613.0	USA
8	Hess Corp	2008	41,165.0	USA
9	Occidental Petroleum Corp	2022	36,634.0	USA
10	Cheniere Energy Inc	2022	33,428.0	USA

Result. The list is led by integrated producers (ConocoPhillips, ENI, Marathon Oil, Hess) rather than pure drillers: SIC 13 classification places some integrated, midstream (Enterprise Products, Cheniere), oilfield-service (SLB) and energy-trading (Dynegy) firms in the sector, and their revenue includes refining, trading and services. Most peaks fall in high-price years (2008, 2011–2014 and 2022). Ranking firms rather than firm-years matters: ranked by firm-year, ConocoPhillips alone would fill 6 of the 10 places.

B.3 Geographical distribution of firms

We kept one row per firm (`drop_duplicates('gvkey')`) and counted unique firms per `location` code.

Code 1.4 · Number of firms per location

```
# Count firms, not firm-years; each gvkey has a single location in this file
firm_locations = df_og.drop_duplicates("gvkey")["location"]
location_counts = firm_locations.value_counts()
top_locations = location_counts.head(10)

# Flag ties at the cut-off so no location is silently dropped
cutoff = location_counts.iloc[9]
tied = location_counts[location_counts == cutoff].index.tolist()
```

Table 1.4. Top 10 firm locations (Hong Kong also has 3 firms, so 10th place is shared with Singapore, Norway and Switzerland).

#	Location	Country	Firms	Share of 772
1	USA	United States	654	84.7%
2	CAN	Canada	54	7.0%
3	GBR	United Kingdom	9	1.2%
4	AUS	Australia	8	1.0%
5	BMU	Bermuda	7	0.9%
6	FRA	France	4	0.5%
7	CYM	Cayman Islands	4	0.5%
8	SGP	Singapore	3	0.4%
9	NOR	Norway	3	0.4%
10	CHE	Switzerland	3	0.4%

Result. The sector is overwhelmingly North American: 84.7% of firms are located in the USA and 7.0% in Canada. The rest is a long tail of 23 countries. Bermuda and the Cayman Islands rank high because offshore drilling contractors often incorporate there.

B.4 Average stock price across the years

We grouped by fiscal year and computed the mean `prcc_c`. Because a few very high-priced shares can pull the average up, we also plotted the median, and marked the 2008 crisis, the 2014–16 oil crash and COVID.

Code 1.5 · Mean and median stock price per year (plotting code shortened)

```
price_by_year = df_og.groupby("fyear")["prcc_c"].agg(["mean", "median"])

fig, ax = plt.subplots(figsize=(9, 4.2))
ax.plot(price_by_year.index, price_by_year["mean"], color=BLUE, lw=2, label="Mean")
ax.plot(price_by_year.index, price_by_year["median"], color=GREY, lw=2, ls="--",
        label="Median")
for yr, label in [(2008, "2008 crisis"), (2015, "2014-16 oil crash"), (2020, "COVID")]:
    ax.annotate(label, (yr, price_by_year.loc[yr, "mean"]), xytext=(0, -22),
                textcoords="offset points", ha="center")
ax.set_ylim(0, None)
```

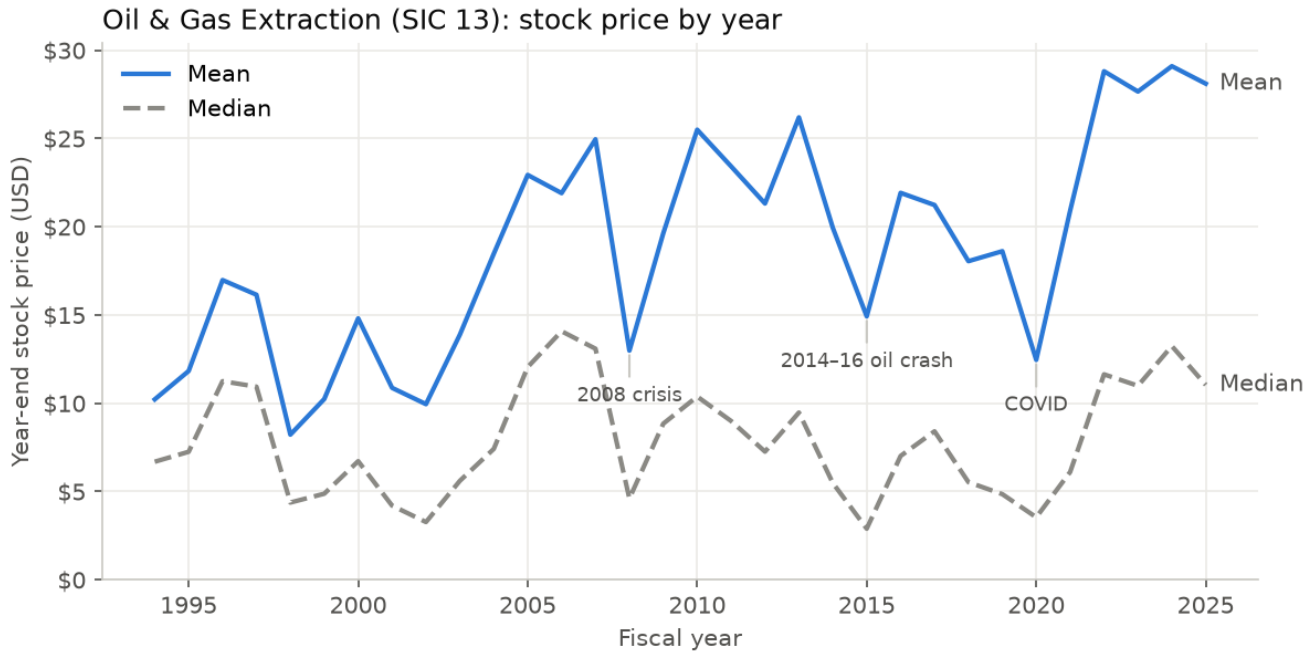


Figure 1.2. Average (mean) and median year-end stock price of SIC 13 firms, 1994–2025.

Result. The mean price tracks the oil cycle: it rises through the 2000s commodity boom to $24.96 in 2007, almost halves to $12.98 in 2008 (−48%), falls again in the 2015 price collapse, reaches its low of $12.46 in 2020 (COVID) and then climbs to a peak of $29.10 in 2024 ($28.11 in 2025). The median is far lower throughout and has not regained its 2006–2007 level ($13.09 in 2007 vs. $11.04 in 2025): the post-2021 rise in the mean is driven by a small number of high-priced survivors rather than by the typical firm.

B.5 Firm most affected by the 2008 financial crisis

We put each firm's 2007 and 2008 prices side by side (one row per firm), kept the 236 firms with a positive price in both years, and computed the percentage change. Because tiny prices create near-100% "drops," we repeated the ranking for stocks priced at $5 or more in 2007.

Code 1.6 · Percentage change in stock price, 2007 → 2008

```
price_07_08 = (df_og[df_og["fyear"].isin([2007, 2008])]
               .pivot_table(index=["gvkey", "conm"], columns="fyear", values="prcc_c")
               .dropna())                                    # need both years
price_07_08 = price_07_08[price_07_08[2007] > 0]      # no % change off a zero base
price_07_08["pct_change"] = (price_07_08[2008] / price_07_08[2007] - 1) * 100
biggest_drops = price_07_08.sort_values("pct_change").head(10)

# Robustness check: exclude penny stocks, where tiny price moves create ~100% "drops"
robust = price_07_08[price_07_08[2007] >= 5].sort_values("pct_change").head(5)
```

Table 1.5. Firm with the largest percentage drop in stock price from 2007 to 2008.

Firm	Price 2007 ($)	Price 2008 ($)	Change
Black Raven Energy Inc	**0.28**	**0.0001**	**−99.96%**
For context: Edge Petroleum Corp (largest drop among firms priced ≥ $5 in 2007)	5.93	0.16	−97.3%

Result. Black Raven Energy was hit hardest, falling from $0.28 to about $0.0001, a 99.96% loss. Because it was already a penny stock, we also checked firms trading at $5 or more in 2007: Edge Petroleum fell 97.3%, followed by South Texas Oil (−94.2%) and PostRock Energy (−93.9%). The crisis hit the whole sector: 92.4% of the 236 firms lost value and the median firm lost 61%, as oil fell from about $140 a barrel in mid-2008 to under $40 by year-end.

B.6 Average return on assets (ROA) of US-located firms

ROA was computed as `net_income / asset` for firms with `location == 'USA'`, excluding rows with zero or missing assets and missing net income. The simple (equal-weighted) mean ROA is not plotted: firms with near-zero assets dominate it (the worst firm-year is −3,268×) and it swings from −1,517% in 2015 to +527% in 2019. The chart shows two averages that hold up: the asset-weighted mean (sector net income divided by sector assets) and the median firm.

Code 1.7 · Average ROA of US-located firms (plotting code shortened)

```
df_usa = (df_og[(df_og["location"] == "USA") & (df_og["asset"] > 0)]
          .dropna(subset=["net_income"]).copy())
df_usa["roa"] = df_usa["net_income"] / df_usa["asset"]

# The equal-weighted mean is dominated by near-zero-asset shells (worst firm-year: -3,268x),
# so we plot the asset-weighted mean (sector net income / sector assets) and the median firm
roa_by_year = df_usa.groupby("fyear").agg(
    raw_mean=("roa", "mean"), median=("roa", "median"),
    total_ni=("net_income", "sum"), total_assets=("asset", "sum"), n_firms=("roa", "size"))
roa_by_year["asset_weighted"] = roa_by_year["total_ni"] / roa_by_year["total_assets"]
```

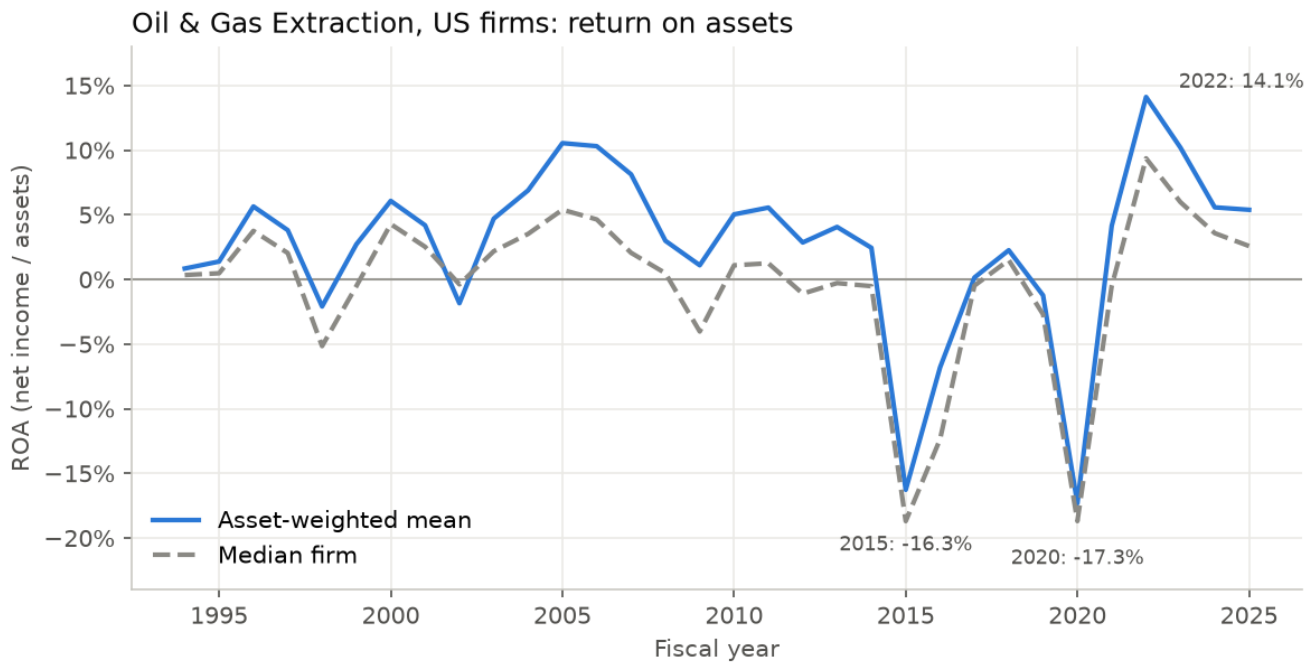


Figure 1.3. Asset-weighted mean and median ROA (net income / assets) of US-located firms, 1994–2025.

Result. The sector earned a positive asset-weighted ROA in most years, peaking at 10.5% in 2005 and 10.3% in 2006, and turned negative in 1998 (−2.1%), 2002 (−1.8%), 2015 (−16.3%), 2016 (−6.8%), 2019 (−1.2%) and 2020 (−17.3%). It rebounded to 14.1% in 2022 and was 5.4% in 2025. The median firm sits much lower than the sector as a whole (2.1% vs. 8.1% in 2007; 2.6% vs. 5.4% in 2025) and was as low as −18.7% in 2015 and 2020: profits are concentrated in the largest firms, while the typical small producer roughly breaks even in good years and loses heavily in downturns. The number of US firms with usable data fell from 243 in 1994 to 105 in 2025.

Part 2 · Text Analysis on the Industry Sector

Oil & Gas Extraction (SIC 13) · Item 1 of 10-K filings, 2020 vs. 2025

> **Pipeline.** C Text cleaning → D Keyword analysis (merge, data-quality screen, top-10 keywords, word clouds) → E Word2Vec embeddings → F 2020 vs. 2025 comparison

C. Text cleaning

The goal of this step was to get the text of the companies' annual reports ready for analysis. The 2020 file of 10-K business descriptions contains 329 exact duplicate rows (same cik, gvkey and text). They were dropped first, because repeated filings would otherwise be counted twice when training Word2Vec, leaving 5,152 of 5,481 rows. Every remaining entry was then cleaned in three steps:

- **Lowercase** all text, so the same word is always counted as the same word however it was capitalized.
- **Remove punctuation**, so "products," and "products" are not treated as different words. Punctuation is replaced with a space, so "oil-and-gas" becomes "oil and gas" rather than "oilandgas"; the pattern also catches Unicode punctuation such as curly quotes and bullets.
- **Remove filler words** such as "the," "and" and "of," using NLTK's standard stop-word list. These words appear in nearly every sentence and say little about the business. Contractions split into fragments ("don't" → "don," "t") that the same list already contains, so they are removed too.

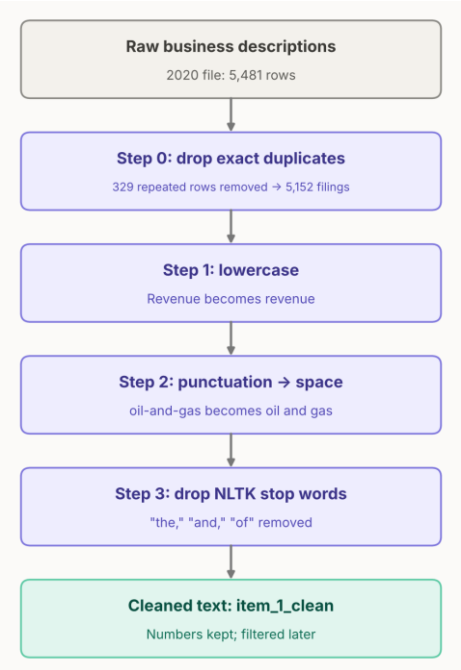


Figure 2.1. Text-cleaning flow chart.

Result. The cleaned column (`item_1_clean`) is lowercase and free of punctuation and common filler words, so each description is shorter and more focused on content: the median filing falls from 6,326 to 4,063 tokens, and 36.3% of all tokens are removed. Numbers are kept at this stage because the cleaning steps do not remove them; they are filtered out later, when building keywords (D) and Word2Vec sentences (E).

Table 2.1. Company name and cleaned text (item_1_clean), first rows of the 2020 file (text truncated).

Firm (name)	Cleaned text (item_1_clean)
ACCESS-POWER INC	fixed expenses previosuly documented 8k 2 350 …
HURCO COMPANIES INC	general hurco companies inc international indu…
ACRO BIOMEDICAL CO., LTD.	engaged business developing marketing products…
Chun Can Capital Group	corporate history chun capital group formerly …
Adaiah Distribution Inc	general adaiah distribution inc incorporated s…

Code 2.1 · Clean Item 1 text: de-duplicate, lowercase, punctuation to space, drop stop words

```
STOP_WORDS = set(stopwords.words("english"))
PUNCT_RE = re.compile(r"[^\w\s]|_")     # also catches Unicode punctuation

def clean_text(text):
    text = text.lower()
    # Replace punctuation with a space so "oil-and-gas" -> "oil and gas", not "oilandgas"
    text = PUNCT_RE.sub(" ", text)
    return " ".join(w for w in text.split() if w not in STOP_WORDS)

def load_and_clean(path):
    df = pd.read_csv(path)
    # 329 exact duplicate rows in the 2020 file would be double-counted in word2vec
    df = df.drop_duplicates().reset_index(drop=True)
    df["item_1_clean"] = df["item_1"].fillna("").map(clean_text)
    return df

df_2020 = load_and_clean("2020_10K_item1_full.csv")     # 5,152 rows
```

D. Keyword analysis

D.1 Merge: one row per firm, keyed on gvkey

The goal was a single table with exactly one row per company, limited to one industry, pairing each firm's business description (10-K file) with its company details (public_firms.csv: industry code, fiscal year, financials). This lets later steps ask how firms in the same industry describe themselves.

- **Choose the industry.** Filter to firms in major group 13 (`sic // 100`, oil and gas extraction).
- **One row per firm, company file.** `public_firms.csv` has one row per firm-year, so for each firm we keep the fiscal year closest to 2020 (the year of the 10-K filings). 129 of the 159 merged firms have a 2020 row; the others use their nearest year.
- **One row per firm, 10-K file.** Where a firm has several entries, keep the one with the longest business description, assuming it is the most complete.
- **Combine.** Inner join on gvkey (firms in both files), giving 159 firms with one row each.
- **Data-quality screen.** After the merge, about 1 in 4 oil and gas rows in the 2020 file contain another company's Item 1 text: ConocoPhillips' row is SunPower's business description, Marathon Oil's is Hershey's and Hess's is CVR Energy's, which would give such firms keywords like "solar" or "chocolate." We flag a filing when the distinctive word in the firm's name never appears in its own text, and we read every flagged row, plus every row with fewer than 10 oil-and-gas words per 1,000, by hand. This dropped 40 filings (saved to `d_dropped_mismatched_2020.csv`), leaving 119

firms. The name test missed three filings (SM Energy, Range Resources and Select Energy Services) that the manual pass caught, and Patterson-UTI was kept after checking that its text is genuine drilling text that simply never names the firm.

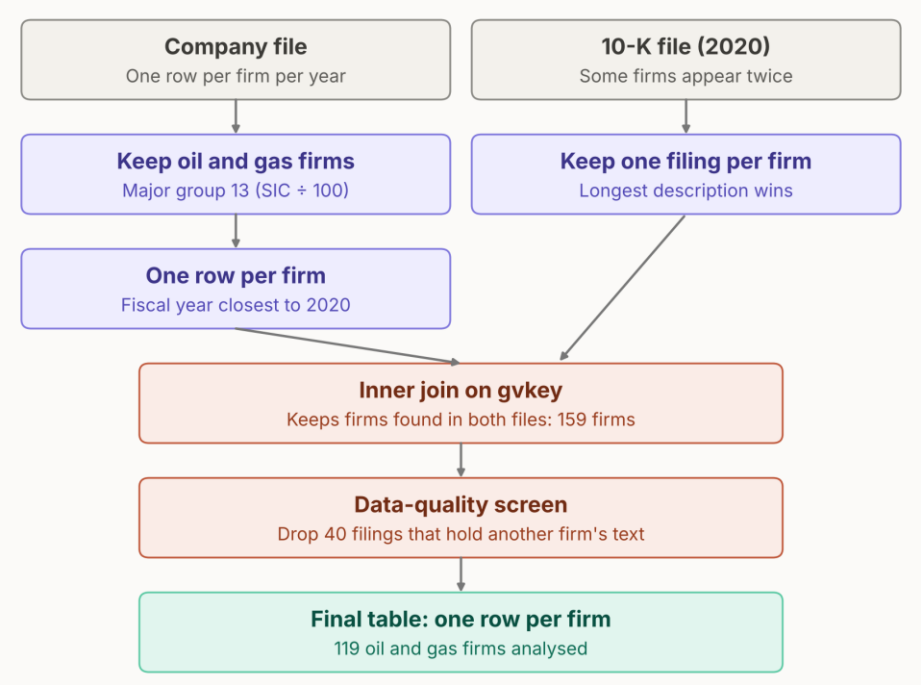


Figure 2.2. Merge flow chart.

Why gvkey? gvkey is a permanent company ID assigned by Compustat. Names and tickers can change after a rebrand, merger or listing change, and two firms can spell the same name differently ("DEVON ENERGY CORP/DE" vs. "DEVON ENERGY CORP"); gvkey stays with a company for its whole life. The other shared ID, cik, is less reliable: in the 10-K file, 285 ciks map to more than one gvkey (for example Arconic).

Result. Each oil and gas company's written description now sits beside its financials from the fiscal year closest to 2020 (sales, assets, location, SIC code). This confirms the two sources were matched correctly and sets up later analysis of how firms describe their business alongside how they perform. The 159 matched firms become 119 after the data-quality screen (Table 2.2).

Table 2.2. Merge and data-quality screen: firm counts for the 2020 sample.

Step	Firms
Oil and gas firms with a 2020 10-K, matched on gvkey	159
of which financials taken from fiscal year 2020	129
Filings dropped by the data-quality screen (another company's text)	40
Firms analysed in Steps D and E	**119**

Code 2.2 · Merge on gvkey, keep one row per firm and screen out mismatched filings

```
YEAR = 2020
df_text = pd.read_pickle(f"df_{YEAR}_clean.pkl")        # output of Part C

# One row per firm: the fiscal year closest to the filing year; merge key = gvkey
sector_firms = public_firms[public_firms["major_group"].isin(SECTOR_CODES)].copy()
sector_firms["year_gap"] = (sector_firms["fyear"] - YEAR).abs()
sector_firms = (sector_firms.sort_values(["gvkey", "year_gap"])
                .drop_duplicates("gvkey")
                [["gvkey", "conm", "sic", "fyear", "sale", "asset", "location"]])
df_og = df_text.merge(sector_firms, on="gvkey", how="inner")      # 159 firms

# A few firms have 2+ different filings; keep the longest
df_og = (df_og.assign(n_chars=df_og["item_1"].str.len())
         .sort_values("n_chars", ascending=False)
         .drop_duplicates("gvkey")
         .drop(columns="n_chars"))

# Data quality: does the firm's distinctive name word appear in its own Item 1?
def name_appears_in_text(name, text):
    words = [w for w in re.findall(r"[a-z]+", name.lower())
             if w not in GENERIC_NAME_WORDS and len(w) > 2]
    return (not words) or (words[0] in text.lower())

df_og["text_matches_firm"] = [
    (name_appears_in_text(n, t) or n in MANUALLY_VERIFIED_OK)
    and n not in MANUALLY_VERIFIED_BAD
    for n, t in zip(df_og["name"], df_og["item_1"])]
df_og = df_og[df_og["text_matches_firm"]].reset_index(drop=True)   # 119 firms
```

D.2 Top-10 keywords per firm (word counts and TF-IDF)

Each firm's long description was boiled down to 10 keywords by two methods that answer different questions: word counts show what a firm talks about most; TF-IDF shows what makes a firm sound different from others in its industry.

- **Prepare.** Tokens are letters-only words of two or more characters (`[a-z][a-z]+`), which removes the numbers and stray single letters (such as "u s") that Step C deliberately kept.
- **Method 1, word counts.** Count every word in the cleaned description (scikit-learn's `CountVectorizer`, fitted on the 119 sector filings) and keep the 10 most frequent.
- **Method 2, TF-IDF.** A word scores high when it is frequent in one firm's description but rare elsewhere. The inverse document frequency is learned on the full 2020 10-K corpus (all industries), so a word scores high when it is frequent for the firm and rare across public companies: the keywords show what sets oil and gas firms apart from other industries. Words found in fewer than 5 filings (`min_df=5`) were ignored so that one-off typos do not earn the maximum score. The 10 highest scores per firm were kept.
- **Save and check.** Both lists are stored as new columns (and saved to `d_keywords_by_firm_2020.csv`); the first rows were printed to confirm they look reasonable.

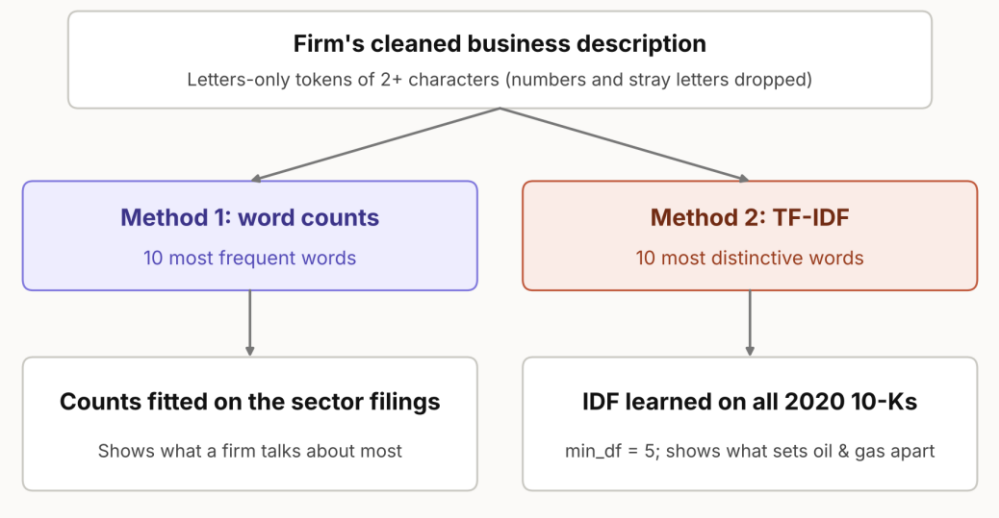


Figure 2.3. Keyword-extraction flow chart.

Result. For most firms both lists are dominated by core industry words such as “oil,” “gas,” “natural” and “wells,” with the TF-IDF list swapping in more specific terms such as “fracturing,” “basin,” “appalachia,” “bcfe” or “offshore” (and place names such as “mauritania” and “senegal” for Kosmos Energy) that help tell firms apart.

Table 2.3. Top-10 keywords by word count and TF-IDF, first six firms (lists truncated; the full table is saved to d_keywords_by_firm_2020.csv).

Firm	Top 10 by word count	Top 10 by TF-IDF
WPX ENERGY INC	may, gas, oil, natural, could, operations, res…	felix, gas, oil, fracturing, reserves, natural…
MATADOR RESOURCES CO	oil, gas, natural, production, december, wells…	oil, gas, natural, mateo, proved, wells, bbl, …
CHORD ENERGY CORP	gas, oil, natural, crude, production, wells, b…	crude, gas, oil, natural, basin, wells, omp, w…
DIAMONDBACK ENERGY INC	oil, gas, natural, wells, production, basin, m…	oil, gas, natural, wells, basin, drilling, per…
MAMMOTH ENERGY SERVICES INC	services, drilling, oil, natural, operations, …	drilling, fracturing, sand, oil, services, unc…
KOSMOS ENERGY LTD	oil, gas, production, well, exploration, devel…	mauritania, senegal, exploration, offshore, pr…

Code 2.3 · Extract top-10 keywords per firm (word counts and TF-IDF)

```
TOKEN_PATTERN = r"(?u)\b[a-z][a-z]+\b"      # letters-only tokens of 2+ characters

count_vec = CountVectorizer(token_pattern=TOKEN_PATTERN)
count_matrix = count_vec.fit_transform(df_og["item_1_clean"])

# IDF learned on the FULL 10-K corpus (all industries); min_df=5 drops one-off typos
tfidf_vec = TfidfVectorizer(token_pattern=TOKEN_PATTERN, min_df=5)
tfidf_vec.fit(df_text["item_1_clean"].drop_duplicates())
tfidf_matrix = tfidf_vec.transform(df_og["item_1_clean"])

def top_n_terms(matrix, vocab, n=10):
    rows = []
    for i in range(matrix.shape[0]):
        scores = matrix[i].toarray().ravel()
        top_idx = scores.argsort()[::-1][:n]
        rows.append([(vocab[j], scores[j]) for j in top_idx if scores[j] > 0])
    return rows

df_og["top10_count"] = top_n_terms(count_matrix, count_vec.get_feature_names_out())
df_og["top10_tfidf"] = top_n_terms(tfidf_matrix, tfidf_vec.get_feature_names_out())
```

D.3 Word clouds

To see the keywords across the whole sector at once, two word clouds were built from the firms' top-10 keywords. Each word's weight is its D.2 score summed over every firm whose top 10 it appears in, so the clouds show exactly the D.2 keywords, sized by how strongly and how widely they recur. The left cloud uses word counts and the right uses TF-IDF scores; a larger word carries more weight across all firms.

What the clouds show. Both are dominated by "gas," "oil" and "natural," confirming the data centers on the oil and gas industry and that both methods capture the sector's core vocabulary. The differences are in the middle and lower ranks:

- **Word counts** surface generic business and reporting words alongside industry terms: "company," "operations," "services," "may," "regulations," "customers" and "December." They are frequent because every annual report uses them, not because they say anything specific about oil and gas.
- **TF-IDF** pushes those words down and brings forward technical terms: "drilling," "wells," "proved," "fracturing," "reserves," "hydraulic," "rigs," "basin," "subsea," "seismic" and "offshore." This reads more like a description of what oil and gas firms do, from drilling methods to reserve classifications.

2020 results. The word-count cloud is led by "gas," "oil," "natural," "production" and "operations"; the TF-IDF cloud by "oil," "gas," "drilling," "natural" and "wells," with more specific terms such as "proved," "fracturing" and "subsea" behind them. "drilling" is a top-10 TF-IDF term for 54 firms and "reserves" for 26.

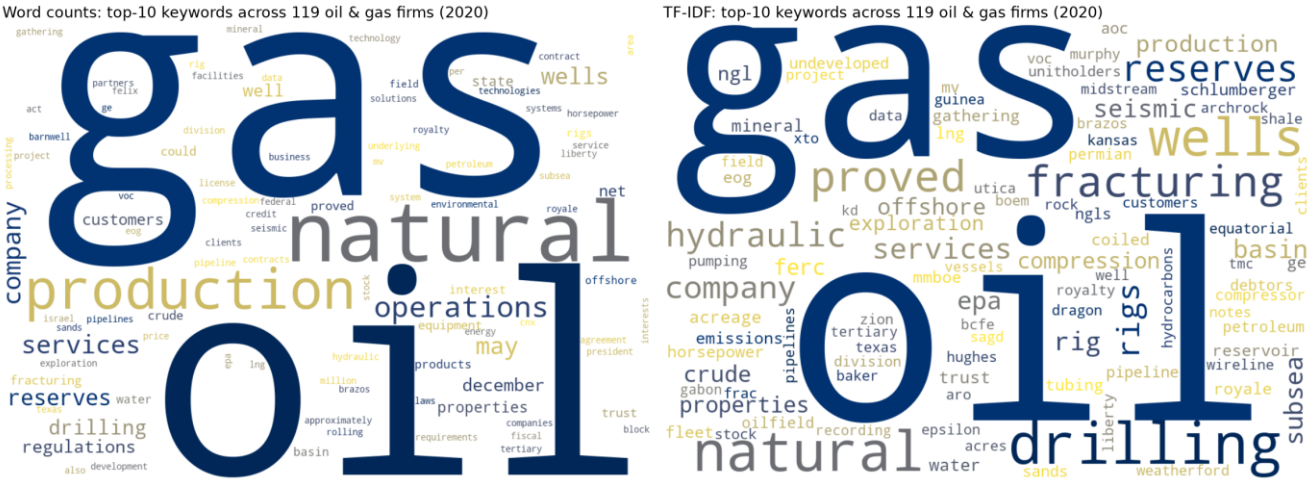


Figure 2.4. 2020 word clouds, 119 firms: word counts (left) and TF-IDF scores (right).

Code 2.4 · Aggregate keyword scores and build word clouds (2020)

```
# Weight = each keyword's score summed over every firm whose top 10 it appears in
def aggregate_keywords(keyword_lists):
    totals = {}
    for kw in keyword_lists:
        for word, score in kw:
            totals[word] = totals.get(word, 0) + score
    return pd.Series(totals).sort_values(ascending=False)

sector_counts = aggregate_keywords(df_og["top10_count"])
sector_tfidf = aggregate_keywords(df_og["top10_tfidf"])

fig, axes = plt.subplots(1, 2, figsize=(16, 6.5))
for ax, weights, title in [(axes[0], sector_counts, "Word counts"),
                           (axes[1], sector_tfidf, "TF-IDF")]:
    wc = WordCloud(width=1000, height=700, background_color="white", colormap="cividis",
                   max_words=100, random_state=42, prefer_horizontal=0.95)
    ax.imshow(wc.generate_from_frequencies(weights.to_dict()), interpolation="bilinear")
    ax.set_title(f"{title}: top-10 keywords across {len(df_og)} oil & gas firms ({YEAR})")
    ax.axis("off")
```

E. Word embeddings: training a Word2Vec model

E.1 Train a Word2Vec model on the full 10-K sample

The goal was a model that learns how words are used in annual reports. Word2Vec gives each word a short list of numbers so that words used in similar ways get similar numbers, which later steps can use to measure how related two words, or two firms, are.

- **Full sample.** The model was trained on the cleaned descriptions of all 5,152 filings in the 2020 10-K file (5,481 rows less 329 exact duplicates), not just oil and gas firms, so it learns from the widest range of business language.
- **Prepare text.** Bare numbers were dropped, and each cleaned description was cut into 1,000-word chunks, because gensim silently ignores every token after the first 10,000 in a "sentence" and Item 1 sections run to 50,000+ tokens. Step C removed punctuation, so real sentences cannot be recovered. Identical texts were used once. This gives 24,473 chunks and 22.2 million tokens.
- **Train.** Word2Vec reads every chunk and learns which words tend to appear near each other; each word ends up with 100 numbers that capture how it is used. We use skip-gram (`sg=1`), which handles less frequent words such as "emissions" better than CBOW, a 5-word window, `min_count=5` and 5 epochs. The vocabulary has 55,232 words.

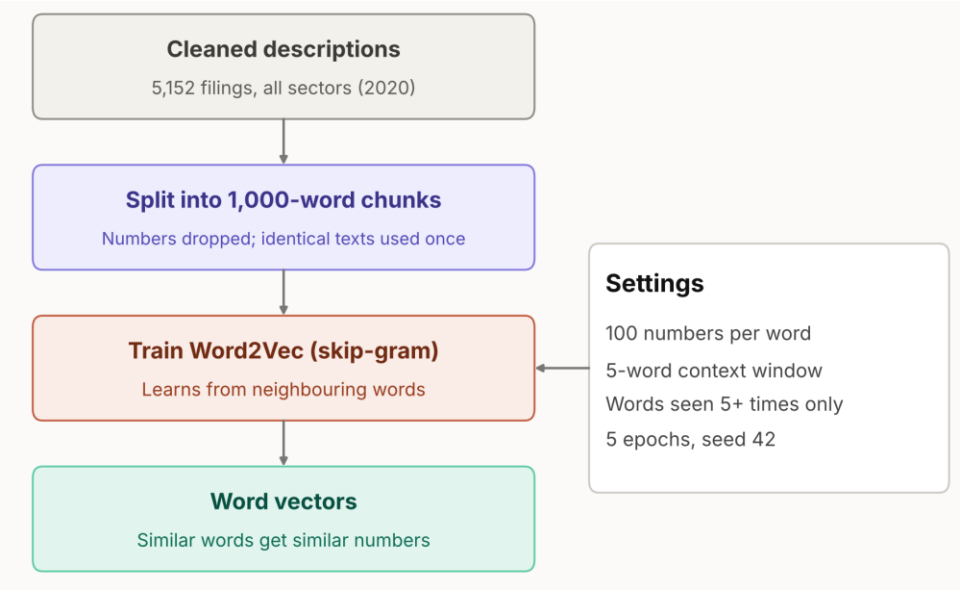


Figure 2.5. Word2Vec training flow chart.

Code 2.5 · Train Word2Vec on the full cleaned 2020 sample

```
CHUNK_SIZE = 1000

def build_sentences(df):
    texts = df["item_1_clean"].drop_duplicates()        # identical texts used once
    sentences = []
    for text in texts:
        tokens = [t for t in text.split() if not t.isdigit()]   # drop bare numbers
        # gensim ignores tokens past 10,000 per "sentence": use 1,000-token chunks
        sentences += [tokens[i:i + CHUNK_SIZE] for i in range(0, len(tokens), CHUNK_SIZE)]
    return sentences

def train_word2vec(df):
    return Word2Vec(build_sentences(df), vector_size=100, window=5, min_count=5,
                    sg=1, epochs=5, seed=42, workers=4)

df_text = pd.read_pickle("df_2020_clean.pkl")            # output of Part C (all sectors)
w2v_model = train_word2vec(df_text)
```

E.2 Representative keywords and their nearest neighbors

Three representative keywords were chosen from the TF-IDF word cloud, one for each side of the business: “drilling” (operations; a top-10 TF-IDF term for 54 firms), “reserves” (the asset base that underpins valuation; 26 firms) and “emissions” (environmental regulation). Table 2.4 shows each word’s five nearest neighbours; a cosine similarity of 1.0 would mean identical usage.

Result. Without being told anything about the industry, the model picked up how oil and gas talks: “drilling” sits next to “workover,” “drilled” and “wells”; “reserves” next to reserve-category jargon (“proved,” “puds,” “mboe”); and “emissions” next to “ghg,” “greenhouse” and “nox” (cosine up to 0.933). Beyond the top five, “emissions” is also close to “methane” (0.835), while “emissions” and “targets” (0.324) or “capture” (0.325), “reserves” and “shareholders” (0.297), and “drilling” and “discipline” (0.199) are not close. **Caveat:** gensim printed `Exception ignored in 'our_dot_float'` warnings during training, likely due to a version mismatch with Python 3.14. We could not rule out an effect on training; the neighbours are consistent with domain knowledge but should be read with that caveat.

Table 2.4. 2020 Word2Vec model: five most similar words (cosine similarity) for “drilling,” “reserves” and “emissions.”

Rank	drilling	reserves	emissions
1	workover (0.808)	proved (0.828)	ghg (0.933)
2	drilled (0.797)	ibnr (0.744)	emission (0.898)
3	wells (0.796)	puds (0.732)	greenhouse (0.886)
4	drill (0.793)	pud (0.731)	ghgs (0.857)
5	exploration (0.777)	mboe (0.727)	nox (0.836)

F. Comparative analysis: 2020 vs. 2025

This step repeats the analysis on oil and gas companies’ 2025 annual reports to see whether the language these firms use to describe their business has changed, looking at the words that stand out in the descriptions and at which words the analysis treats as closely related.

- **Collect** the 2025 business descriptions for the same oil and gas industry used in 2020.
- **Clean** the text the same way: remove exact duplicates, lowercase, turn punctuation into spaces and drop filler words.
- **Find the most important words** two ways: by how often a word appears, and by words that are frequent in one company’s description but uncommon in others’, which shows what makes a company different.
- **Draw a word cloud** for each method; the bigger the word, the more weight it carries across all companies.
- **Train a language model** for each year that learns which words tend to be used together, then ask which words it sees as closest to “drilling,” “reserves” and “emissions.”
- **Compare everything with 2020**, using the share of firms that have each word in their top-10 TF-IDF keywords (summed TF-IDF scores are not comparable across years) and which words are neighbours (cosine scores from separately trained models are not comparable).

Both years go through the same code. The merge keeps the fiscal year closest to each filing year, and the mismatch screen now also drops filings of fewer than 200 words (stubs that only say “incorporated by reference”). In 2020, 159 firms merge and 119 are valid; in 2025, 88 merge and 85 are valid.

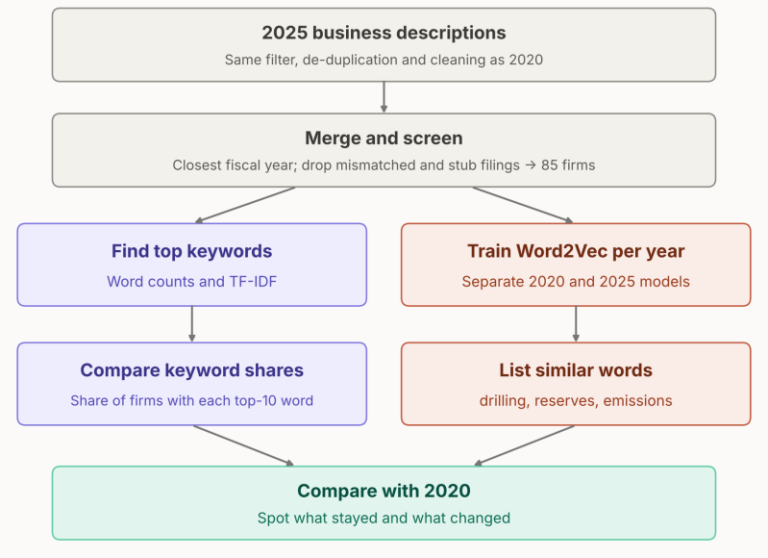


Figure 2.6. 2025 analysis flow chart.

F.1 Word clouds, 2020 vs. 2025

Figure 2.7 shows both years side by side. The 2025 clouds are dominated by “gas,” “oil,” “natural,” “production” and “operations,” confirming that the descriptions center on the oil and gas business. The 2025 TF-IDF cloud also elevates more specific terms such as “drilling,” “reserves,” “offshore,” “subsea” and “emissions,” showing what sets individual companies apart beyond the shared industry vocabulary.

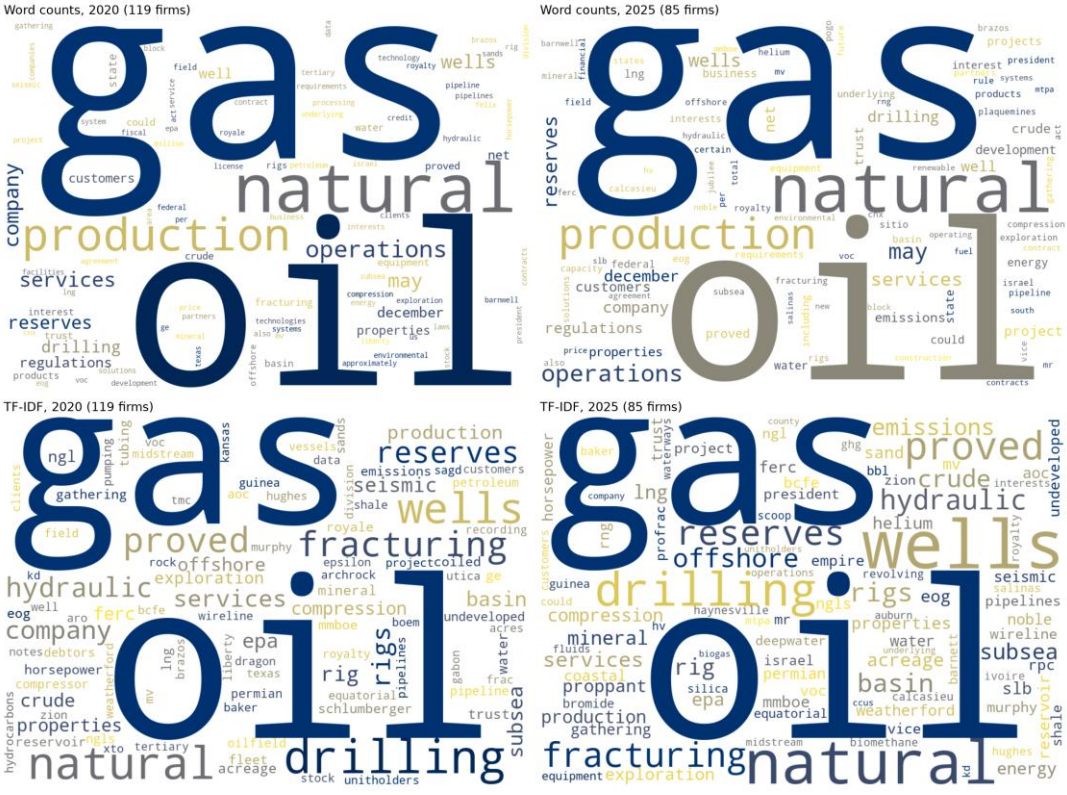


Figure 2.7. Word clouds, 2020 (left, 119 firms) and 2025 (right, 85 firms): word counts (top) and TF-IDF scores (bottom).

F.2 Keyword shares: what moved

Summed TF-IDF scores are not comparable across years (the firm counts differ and the IDF is fitted per year), so we compare the share of firms that have each word in their top-10 TF-IDF keywords. Table 2.5 shows the eight biggest falls and rises.

Table 2.5. Share of firms with each word in their top-10 TF-IDF keywords, 2020 (119 firms) vs. 2025 (85 firms); change in percentage points.

Falls	2020	2025	Change	Rises	2020	2025	Change
oil	87.4%	76.5%	−10.9	reserves	21.8%	30.6%	+8.7
company	12.6%	2.4%	−10.3	emissions	5.9%	14.1%	+8.2
drilling	45.4%	35.3%	−10.1	offshore	7.6%	14.1%	+6.6
coiled	4.2%	0.0%	−4.2	proppant	0.0%	4.7%	+4.7
ferc	8.4%	4.7%	−3.7	proved	28.6%	32.9%	+4.4
fleet	5.9%	2.4%	−3.5	crude	5.0%	9.4%	+4.4
well	3.4%	0.0%	−3.4	energy	1.7%	5.9%	+4.2
ngl	6.7%	3.5%	−3.2	sand	0.8%	4.7%	+3.9

F.3 Word2Vec neighbours, 2020 vs. 2025

The two models were trained separately, so their cosine scores are on different scales; we compare which words are neighbours, not the scores. Table 2.6 lists the ten nearest neighbours of “drilling,” “reserves” and “emissions” in each year.

Table 2.6. Word2Vec nearest neighbours (cosine similarity), 2020 vs. 2025.

#	drilling 2020	drilling 2025	reserves 2020	reserves 2025	emissions 2020	emissions 2025
1	drilled (0.80)	drill (0.79)	proved (0.81)	proved (0.81)	ghg (0.93)	ghg (0.94)
2	drill (0.80)	exploration (0.78)	ibnr (0.75)	pdnp (0.75)	emission (0.89)	greenhouse (0.89)
3	workover (0.80)	recompletion (0.77)	pud (0.75)	ibnr (0.73)	greenhouse (0.88)	emission (0.87)
4	wells (0.79)	wells (0.76)	puds (0.74)	tcfe (0.73)	ghgs (0.86)	methane (0.86)
5	exploration (0.78)	completions (0.76)	unproved (0.73)	undrilled (0.72)	nox (0.83)	ghgs (0.84)
6	rig (0.76)	workover (0.76)	mboe (0.72)	undeveloped (0.72)	methane (0.82)	dioxide (0.82)
7	wellbore (0.76)	workovers (0.73)	undiscounted (0.72)	recoverable (0.71)	tailpipe (0.80)	co2 (0.82)
8	recompletion (0.76)	acidizing (0.73)	undeveloped (0.72)	pud (0.71)	dioxide (0.80)	carbon (0.80)
9	horizontal (0.75)	drilled (0.73)	lae (0.71)	mboe (0.70)	pollutants (0.79)	tailpipe (0.80)
10	completions (0.75)	wellbore (0.72)	recoverable (0.71)	mmcfe (0.70)	so2 (0.78)	nox (0.79)

F.4 What changed between 2020 and 2025, and why

1. Less about drilling, more about reserves. "drilling" is a top-10 TF-IDF term for 45% of firms in 2020 and 35% in 2025, and "oil" falls from 87% to 77%. Meanwhile "reserves" rises from 22% to 31% and "proved" from 29% to 33%. After the 2020 price collapse, producers tended to shift from growth through drilling to capital discipline and returns to shareholders. Their filings put more emphasis on the value of the reserves they already hold than on drilling activity.

2. Emissions became a mainstream topic. With the increasing awareness about climate action, "emissions" grows from 6% to 14% of firms. In Word2Vec, "methane" moves from 6th to 4th among the closest words, and "co2" and "carbon" enter the top 10 in 2025. Stricter methane regulation and investor and regulator demands for climate disclosure likely pushed emissions language into the business description itself.

3. Completion technology. "proppant" (0% moved to → 5%) and "sand" (1% → 5%) appear among top keywords. In Word2Vec, "recompletion" (8th → 3rd) and "completions" (10th → 5th) move closer to "drilling." This is consistent with the industry getting more output from each well through completion design.

Caveats. The sample shrinks from 119 to 85 firms, partly through mergers and firms leaving the market, so some of these shifts reflect which firms are included rather than changes in what firms say. The two Word2Vec models were trained separately, so we compare which words are neighbours, not the similarity scores.

Appendix to Part 2: 2020 vs. 2025 code

Code 2.6 · Clean and screen both years with the same code

```
YEARS = [2020, 2025]

def clean_text(text):
    text = re.sub(r"[^\w\s]|_", " ", text.lower())
    return " ".join(w for w in text.split() if w not in STOP_WORDS)

df_text = {}
for year in YEARS:
    df = pd.read_csv(f"{year}_10K_item1_full.csv").drop_duplicates()   # 2020: 329 dupes
    df["item_1_clean"] = df["item_1"].fillna("").map(clean_text)
    df_text[year] = df

def is_valid_filing(year, name, text):
    # <200 words = "incorporated by reference" stub
    if len(text.split()) < 200 or (year, name) in MISLABELLED:
        return False
    if (year, name) in GENUINE:
        return True
    name_words = [w for w in re.findall(r"[a-z]+", name.lower())
                  if w not in GENERIC_WORDS and len(w) > 2]
    return not name_words or name_words[0] in text.lower()

df_og = {}
for year in YEARS:
    # one row per firm: the firm-year closest to the filing year; merge key = gvkey
    firms = public_firms[public_firms["major_group"] == 13].copy()
    firms["gap"] = (firms["fyear"] - year).abs()
    firms = (firms.sort_values(["gvkey", "gap"]).drop_duplicates("gvkey")
             [["gvkey", "conm", "sic"]])
    df = df_text[year].merge(firms, on="gvkey", how="inner")
    df = (df.loc[df["item_1"].str.len().sort_values(ascending=False).index]
          .drop_duplicates("gvkey"))
    valid = [is_valid_filing(year, n, t) for n, t in zip(df["name"], df["item_1"])]
    df_og[year] = df[valid].reset_index(drop=True)     # 2020: 159 -> 119; 2025: 88 -> 85
```

Code 2.7 · Top-10 keywords and keyword shares, 2020 vs. 2025

```
# (inside the same loop over YEARS: letters-only tokens, IDF learned on all industries)
```

```
token_pattern = r"(?u)\b[a-z][a-z]+\b"
count_vec = CountVectorizer(token_pattern=token_pattern)
count_matrix = count_vec.fit_transform(df["item_1_clean"])
tfidf_vec = TfidfVectorizer(token_pattern=token_pattern, min_df=5).fit(
    df_text[year]["item_1_clean"].drop_duplicates())
tfidf_matrix = tfidf_vec.transform(df["item_1_clean"])
df["top10_count"] = top_10(count_matrix, count_vec.get_feature_names_out())
df["top10_tfidf"] = top_10(tfidf_matrix, tfidf_vec.get_feature_names_out())

# Summed TF-IDF isn't comparable across years (different firm counts, IDF fit per year),
# so compare the share of firms that have each word in their top 10
def top10_share(df, col):
    return pd.Series([w for kw in df[col] for w, _ in kw]).value_counts() / len(df)

keyword_shift = pd.DataFrame(
    {y: top10_share(df_og[y], "top10_tfidf") for y in YEARS}).fillna(0)
keyword_shift["change"] = keyword_shift[2025] - keyword_shift[2020]
keyword_shift = keyword_shift.sort_values("change")
```

Code 2.8 · Word clouds for both years

```
def keyword_weights(df, col):
    weights = {}
    for kw in df[col]:
        for word, score in kw:
            weights[word] = weights.get(word, 0) + score
    return weights

fig, axes = plt.subplots(2, 2, figsize=(16, 12))
for row, col, label in [(0, "top10_count", "Word counts"), (1, "top10_tfidf", "TF-IDF")]:
    for ax, year in zip(axes[row], YEARS):
        cloud = WordCloud(width=1000, height=700, background_color="white",
                          colormap="cividis",
                          max_words=100, random_state=42).generate_from_frequencies(
                              keyword_weights(df_og[year], col))
        ax.imshow(cloud, interpolation="bilinear")
        ax.set_title(f"{label}, {year} ({len(df_og[year])} firms)")
        ax.axis("off")
```

Code 2.9 · Train one Word2Vec model per year and list nearest neighbours

```
# gensim ignores tokens past 10,000 per sentence: split filings into 1,000-word chunks
def to_sentences(texts, size=1000):
    sentences = []
    for text in texts.drop_duplicates():
        tokens = [t for t in text.split() if not t.isdigit()]
        sentences += [tokens[i:i + size] for i in range(0, len(tokens), size)]
    return sentences

KEYWORDS = ["drilling", "reserves", "emissions"]
w2v_models = {}
for year in YEARS:
    w2v_models[year] = Word2Vec(to_sentences(df_text[year]["item_1_clean"]), vector_size=100,
                                window=5, min_count=5, sg=1, epochs=5, seed=42, workers=4)

# Models are trained separately: compare which words are neighbours, not the scores
for kw in KEYWORDS:
    for year in YEARS:
        print(kw, year, w2v_models[year].wv.most_similar(kw, topn=10))
```

Part 3 · Comprehensive Analysis of One Sample Firm

EOG Resources Inc. (SIC 1311, crude petroleum and natural gas · USA)

> **Pipeline.** G.0 Choose the firm → G.1 Peer group → G.2 Revenue, market share and ROA vs. peers → G.3 Asset growth vs. returns → One recommendation

G. Firm analysis and strategy recommendation

G.0 Choosing the focal firm

Why EOG. EOG Resources had the third-highest 2025 sales ($22.6 billion) among US SIC 1311 firms in the dataset, behind ConocoPhillips and Enterprise Products (a pipeline business); among the oil and gas producers in our peer set it ranks second, after ConocoPhillips. It has also been present in all 32 years of the dataset, which allows a long comparison.

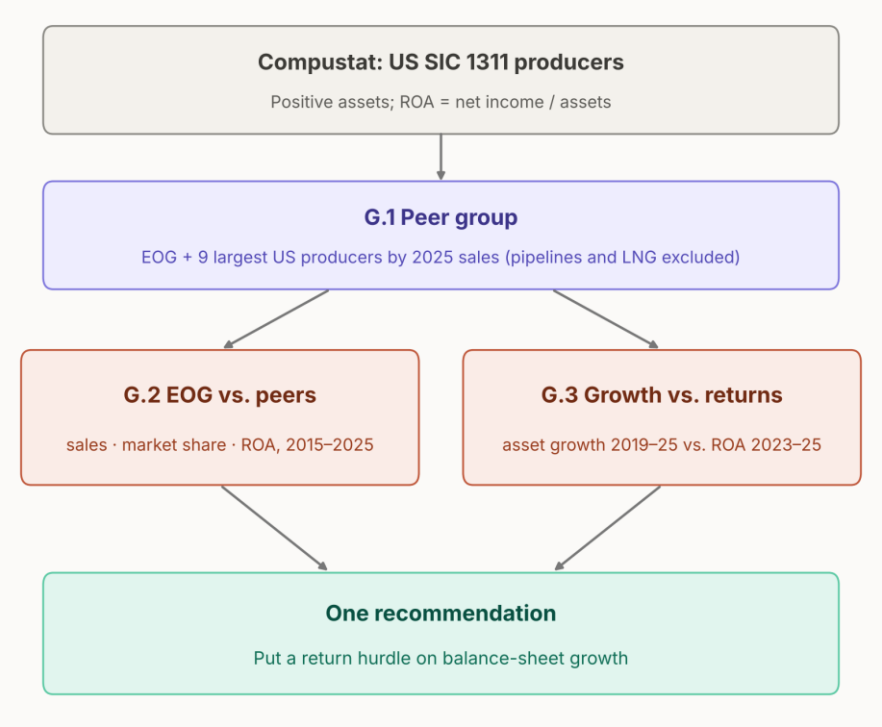


Figure 3.1. Part 3 approach flow chart.

Direction. The guideline asks us to focus on one direction. We compare EOG's **revenue, market share and ROA** with its competitors (the brief's second suggested direction), using the 10 largest US producers as the peer group (G.1). EOG's asset growth against its returns (G.3) is used as supporting evidence for the recommendation.

G.1 Peer group

The goal was a peer group of comparable producers with the same business model as EOG.

- **Universe.** US-located firms with SIC code 1311 (crude petroleum and natural gas) and positive assets; ROA = net income / assets.
- **Peers.** The 10 largest of these by 2025 sales: EOG, ConocoPhillips, Occidental, Devon, Diamondback, Expand Energy, Coterra, APA, Ovintiv and EQT, analysed over fiscal years 2015–2025.

- **Exclusions.** Enterprise Products (pipelines), Cheniere and Venture Global (LNG export) carry SIC 1311 but run different businesses, so they are not peers.
- **Market share.** EOG's sales divided by the total sales of all US SIC 1311 firms in the dataset: a share of listed-producer revenue, not of national oil and gas output.

Code 3.1 · Define the US producer universe and the 10-firm peer group

```
public_firms = pd.read_csv("public_firms.csv")

# Universe for market share: US crude & natural gas producers (SIC 1311)
producers = public_firms[(public_firms["sic"] == 1311) & (public_firms["location"] == "USA")
                         & (public_firms["asset"] > 0)].copy()
producers["roa"] = producers["net_income"] / producers["asset"]

# Peers: the largest US producers by 2025 sales. Enterprise Products (pipelines) and
# Cheniere / Venture Global (LNG export) carry SIC 1311 but run different businesses.
PEERS = {"EOG RESOURCES INC": "EOG", "CONOCOPHILLIPS": "ConocoPhillips",
         "OCCIDENTAL PETROLEUM CORP": "Occidental", "DEVON ENERGY CORP": "Devon",
         "DIAMONDBACK ENERGY INC": "Diamondback", "EXPAND ENERGY CORP": "Expand",
         "COTERRA ENERGY INC": "Coterra", "APA CORP": "APA", "OVINTIV INC": "Ovintiv",
         "EQT CORP": "EQT"}
peers = producers[producers["conm"].isin(PEERS) & (producers["fyear"] >= 2015)].copy()
peers["firm"] = peers["conm"].map(PEERS)
```

G.2 EOG vs. peers: revenue, market share and ROA

For each year from 2015 to 2025 we took EOG's sales, market share and ROA, and compared its ROA with the median ROA of the other nine peers and with their pooled ROA (peer net income divided by peer assets). Table 3.1 shows the results.

Code 3.2 · Sales, market share and ROA of EOG vs. its peers; asset growth vs. returns

```
sales = peers.pivot_table(index="fyear", columns="firm", values="sale")
roa = peers.pivot_table(index="fyear", columns="firm", values="roa")
market_share = sales.div(producers.groupby("fyear")["sale"].sum(), axis=0)

others = peers[peers["firm"] != "EOG"]
summary = pd.DataFrame({
    "EOG sales ($M)": sales["EOG"],
    "EOG market share": market_share["EOG"],
    "EOG ROA": roa["EOG"],
    # median = typical peer; pooled = peer group's total net income / total assets
    "Peer median ROA": others.groupby("fyear")["roa"].median(),
    "Peer pooled ROA": (others.groupby("fyear")["net_income"].sum()
                        / others.groupby("fyear")["asset"].sum()),
    "EOG ROA rank (of 10)": roa.rank(axis=1, ascending=False)["EOG"],
})
summary["EOG premium vs median"] = summary["EOG ROA"] - summary["Peer median ROA"]

growth = pd.DataFrame({
    "Asset growth 2019-25": peers[peers["fyear"] == 2025].set_index("firm")["asset"]
                            / peers[peers["fyear"] == 2019].set_index("firm")["asset"] - 1,
    "Sales growth 2019-25": sales.loc[2025] / sales.loc[2019] - 1,
    "ROA 2025": roa.loc[2025],
    "Avg ROA 2023-25": roa.loc[2023:2025].mean(),
})
```

Table 3.1. EOG vs. its nine peers, 2015–2025 (Compustat; ROA = net income / assets; market share of all US SIC 1311 producers' sales).

Year	EOG sales ($m)	EOG market share	EOG ROA	Peer median ROA	Peer pooled ROA	EOG ROA rank (of 10)	Premium vs. median (pts)
2015	8,718	4.0%	−16.8%	−20.0%	−28.8%	4	+3.2
2016	7,363	4.2%	−3.7%	−6.2%	−6.5%	4	+2.5
2017	11,226	4.7%	8.7%	5.1%	2.8%	1	+3.5
2018	17,011	6.0%	10.1%	8.0%	6.4%	3	+2.1
2019	17,122	6.9%	7.4%	−0.6%	0.8%	3	+8.0
2020	11,018	6.5%	−1.7%	−25.6%	−20.4%	2	+23.9
2021	18,517	6.3%	12.2%	8.9%	8.3%	3	+3.3
2022	25,510	5.4%	18.8%	20.2%	20.0%	7	−1.4
2023	24,000	6.7%	17.3%	10.8%	10.4%	2	+6.5
2024	23,582	7.4%	13.6%	5.0%	4.9%	1	+8.6
2025	22,595	6.9%	9.6%	6.4%	5.2%	1	+3.2

Result. EOG's advantage comes from **returns, not size**. It beat the peer-median ROA in 10 of 11 years (the exception is 2022, when peers were at the top of the cycle: 20.2% vs. EOG's 18.8%), ranked first in 2017, 2024 and 2025, and held up best in the 2020 COVID shock (−1.7% vs. a −25.6% median, a 23.9-point gap). Its share of listed US producers' sales rose from 4.0% in 2015 to 6.9% in 2019 and was 6.9% in 2025 (peak 7.4% in 2024).

The advantage is eroding. EOG's ROA fell from 18.8% in 2022 to 9.6% in 2025. In 2025 its assets grew 9.8% ($47.2 billion to $51.8 billion) while sales fell 4.2% ($23.6 billion to $22.6 billion) and net income fell 22.2% ($6.4 billion to $5.0 billion). Recent capital is therefore not yet earning at EOG's historical level, although commodity prices and acquisition accounting also move year-end ROA (see the caveats).

G.3 Asset growth vs. returns

To test whether growing the balance sheet pays off, we compared each peer's asset growth from 2019 to 2025 with its average ROA over 2023–25 (Table 3.2 and Figure 3.2, right).

Table 3.2. Asset growth, sales growth and ROA of EOG and its peers, 2019–2025 (sorted by asset growth).

Firm	Asset growth 2019–25	Sales growth 2019–25	ROA 2025	Avg ROA 2023–25
Coterra	440.2%	270.0%	7.1%	6.7%
Diamondback	202.0%	279.1%	2.3%	6.0%
Devon	130.4%	176.3%	8.4%	11.0%
EQT	122.2%	95.7%	4.9%	4.1%
Expand	74.7%	43.6%	6.4%	6.9%
ConocoPhillips	72.9%	81.0%	6.6%	8.5%
EOG	**39.5%**	**32.0%**	**9.6%**	**13.5%**
APA	−1.9%	41.2%	8.1%	10.3%
Ovintiv	−5.1%	32.4%	6.1%	7.5%
Occidental	−23.0%	5.9%	2.8%	4.2%

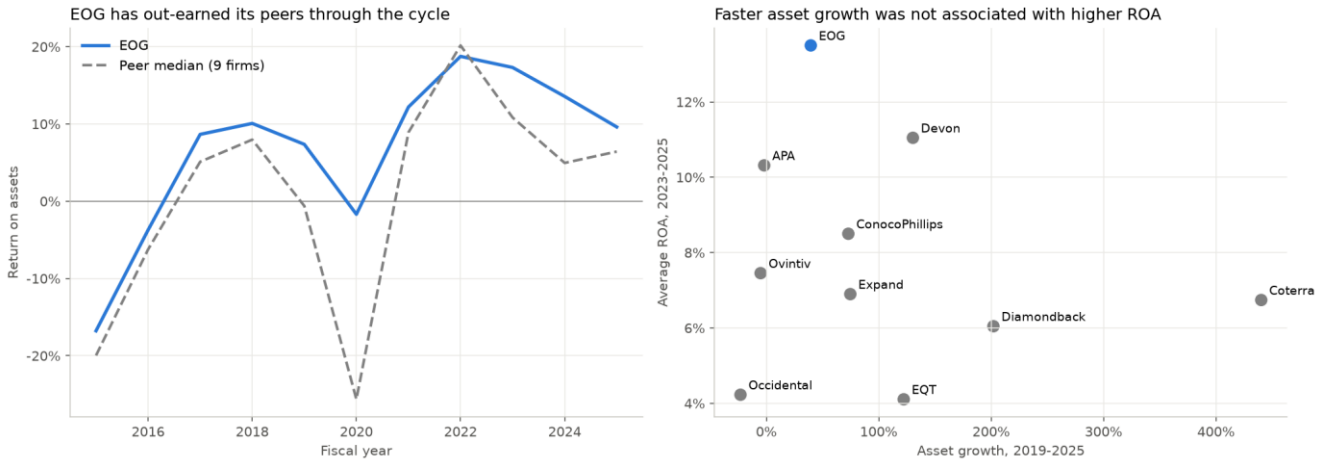


Figure 3.2. Left: EOG vs. peer-median ROA, 2015–2025. Right: asset growth 2019–2025 vs. average ROA 2023–2025 (EOG in blue).

- **Growth did not buy higher returns.** The peers that grew assets fastest (Coterra, Diamondback, EQT) earned no more than slower-growing peers; the correlation between asset growth and average ROA is −0.17 (n = 10).
- **EOG grew assets more slowly than most peers** (+39.5% from 2019 to 2025, seventh of ten) **yet earned the highest average ROA in 2023–25 (13.5%)** and the highest ROA in 2025 (9.6%).
- **Sales did not keep pace with assets in 2025.** EOG's assets rose 9.8% while sales and net income fell, which is why the recommendation below targets how new capital is deployed.

Our recommendation to EOG Resources

Put an explicit return hurdle on balance-sheet growth.

- **What.** New acquisitions and major capital programmes should be expected to earn at least EOG's own 2023–25 average ROA (about 13.5%) before they are approved.
- **Why: returns, not size.** EOG's advantage comes from returns, not size. It beat the peer-median ROA in 10 of 11 years and led the group in 2024–25 while growing assets more slowly than most peers (Tables 3.1 and 3.2).
- **Why: the advantage is eroding.** ROA fell from 18.8% to 9.6% over 2022–25, and in 2025 assets grew 9.8% while sales and net income fell. Recent capital is not yet earning at EOG's historical level.
- **Why: scale has not paid in this group.** Peers that grew assets fastest did not earn more (Coterra, Diamondback, EQT), so chasing scale through acquisitions is not a proven route to better returns.
- **What it protects.** A return hurdle protects the advantage that set EOG apart in the 2020 downturn, and makes clear to investors why recent asset growth should translate into earnings.
- **Caveats.** ROA uses year-end book assets and GAAP net income, so impairments and acquisition accounting can move it sharply. Commodity prices drive most year-to-year swings for every producer. The peer group is small (10 firms) and the growth–ROA relationship is weak (r = −0.17), so this is a judgement supported by the evidence, not a statistical proof.

Conclusions

Taken together, the three parts tell one consistent story about the oil and gas extraction sector:

- **Financially, the sector is cyclical and unforgiving.** Prices and returns collapse in each downturn (2008, 2015, 2020), and only 5% of firms survived the full 32 years. Averages are pulled around by a few large firms, so medians are the better guide to the typical company.
- **Textually, the core vocabulary is stable.** Oil, gas and natural dominate both 2020 and 2025 filings; the changes are at the margin: "drilling" loses share among firms' top keywords, while "reserves," "emissions," offshore work and completion terms such as "proppant" and "sand" gain share. The 2025 sample is smaller (85 vs. 119 firms), so part of this reflects which firms are included.
- **Returns, not size, set the best firm apart.** Among the ten largest US producers, EOG Resources beat the peer-median ROA in 10 of 11 years while growing assets more slowly than most peers, but its ROA has fallen from 18.8% to 9.6% since 2022. That motivates a return hurdle on future balance-sheet growth.

Limitations

- **Text and financial years differ.** A 10-K labelled 2020 mostly describes fiscal 2019, so text and Compustat rows are matched by firm, using the fiscal year closest to the filing year (129 of 159 firms have a 2020 row), rather than by exact fiscal year.
- **Some 10-K text does not belong to the matched firm.** About 1 in 4 oil and gas rows in the supplied 2020 file contained another company's Item 1 text (for example, ConocoPhillips' row described SunPower). We dropped 40 such filings (159 → 119 firms) using a name test and a manual review of low oil-and-gas-vocabulary rows. The 2025 file was screened with the name test and a 200-word minimum only (3 of 88 filings dropped), so some residual noise may remain.
- **Peer group and market share.** The peer group is the 10 largest US SIC 1311 producers by 2025 sales, a small group chosen with hindsight. Market share is a share of all US SIC 1311 firms' sales in the dataset (which still includes pipeline and LNG firms in the denominator), not of national oil and gas output.
- **Book-value ROA and missing variables.** ROA uses year-end book assets and GAAP net income, so impairments and acquisition accounting can move it sharply (for example, a mid-year acquisition raises year-end assets but adds only part-year earnings), and commodity prices drive most year-to-year swings for every producer. The dataset has no capital expenditure, production or project-level data, so the recommendation is a decision framework rather than a project evaluation.
- **Small samples.** The peer group has 10 firms and the growth–ROA relationship is weak ($r = -0.17$), so the recommendation is a judgement supported by the evidence, not a statistical proof. In Part 2 the 2020 and 2025 samples differ (119 vs. 85 firms), the two Word2Vec models were trained separately and gensim printed warnings during training (see E.2), so we compare neighbour words rather than similarity scores. `prcc_c` is not adjusted for splits or reverse splits.

References

[1] BAIT 508 course datasets: public_firms.csv, major_groups.csv, 2020_10K_item1_full.csv and 2025_10K_item1_full.csv (UBC Sauder School of Business). Lecture code provided by the instructor was used as the starting point for text cleaning and Word2Vec training.

[2] Mikolov, T., Sutskever, I., Chen, K., Corrado, G., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. Advances in Neural Information Processing Systems, 26.

[3] Software: pandas, scikit-learn (CountVectorizer, TfidfVectorizer), NLTK (English stop words), gensim (Word2Vec), wordcloud and matplotlib.

Use of generative AI tools

As permitted by the course, team members used generative AI tools. Following the course's "Guidelines for Using Generative AI Tools in Group Projects," each use is attributed below. All numbers in this report come from our notebook outputs and the course datasets, and every AI output was checked by the team member who used it.

Karim Mazrou (Part 1 and report assembly)

Name, version, company	Claude (Opus 5.5), Anthropic, used in the Claude desktop app (Claude Code)
Chat objective	Assemble the final group report in Word from our notebook (BAIT FINAL.ipynb), Anirudha's Part 2 write-up and Madhav's draft part 3 and code check; re-checked project figures; format the report consistently.
Use of AI-generated content	Draft text, tables and the two flow charts (Figures 1.1 and 3.1) were reviewed and edited by the team. The tool re-ran the Part 1 calculations on the course data to confirm the notebook outputs.
Evaluation and limitations	I used Claude code to verify my code, introduce some new useful functions, such as statrswith() and nunique(), as well as using the team's combined observations to generate the final text analysis report. Results were satisfying, though it needed human review for verification.
URL of chat history	https://claude.ai/share/24c92e69-d017-4b37-a86f-f68c6b97cb67
Time and date of chat	9 October 2026, 18.00 pm PST

Anirudha Nair (Part 2 code & text report)

Name, version, company	Claude (Opus 5.5), Anthropic, used in the Claude desktop app
Chat objective	Drafting parts and reviewing over the whole code
Use of AI-generated content	The team compiled different codes into the final code file.
Evaluation and limitations	AI tool was helpful and got the job done, some new functions were introduced
URL of chat history	https://claude.ai/artifact/15rtXCwb39VrKtGoWTGBvn#3450ce07-b5fa
Time and date of chat	9 October 2026, 18.00 pm PST

Madhav Singh (Part 3 code & text report))

Name, version, company	CodeX (GPT 6- Luna), OpenAI
Chat objective	Assistance with structuring and drafting
Use of AI-generated content	The generated code was used as a starting point and modified to fit the project datasets. I reviewed, tested and adjusted the code as needed, then independently interpreted and reported the results
Evaluation and limitations	Codex required continuous course correction to follow project guidelines and frequently produced syntax errors.
URL of chat history	https://chatgpt.com/s/cx_6ac83a5d5e0c8191a07325e6749e0bdc
Time and date of chat	4 October, 2026, Time - 12: 24 PM